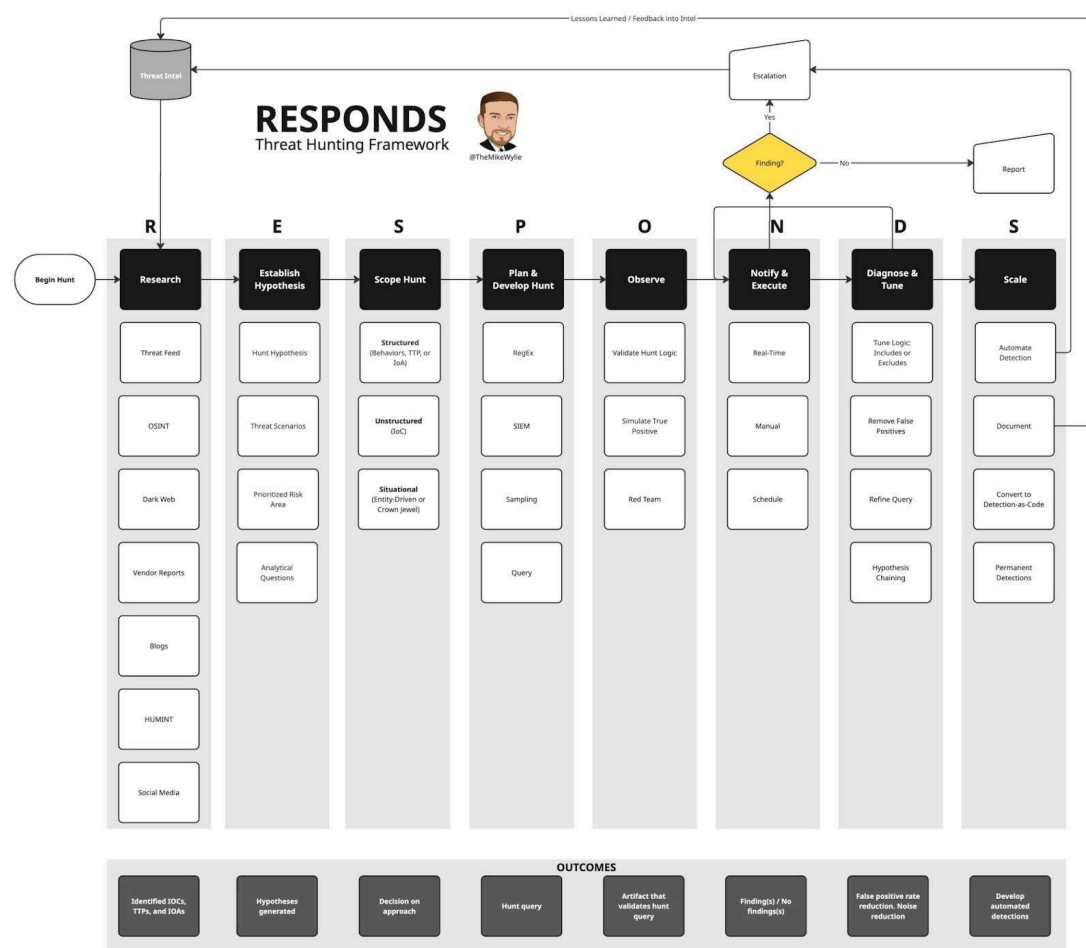


Lab: Mind Mapping Threat Hunting Hypotheses using SWG / ZIA Telemetry

Background

Threat hunting often begins before queries are written.

In mature threat hunting programs, experienced hunters spend time researching, brainstorming, and forming hypotheses based on how attackers commonly abuse infrastructure, protocols, and trusted services. Just as Abraham Lincoln said, "Give me six hours to chop down a tree and I will spend the first four sharpening the axe," we want to plan and scope our hunt to avoid operational traps like days spent chasing poor leads down a rabbit hole.



The Research and Establish Hypothesis steps are critical when hunting in high-volume datasets like Secure Web Gateway (SWG) or Zscaler Internet Access (ZIA) telemetry, where searching without direction quickly becomes unmanageable.

Mind mapping is a simple but powerful technique to:

- Break down a broad problem into manageable components
- Encourage creative and adversary-focused thinking
- Generate multiple hunt hypotheses from a single data artifact
- Help newer threat hunters formulate a hunt plan
- Unblock 'writers block'

In this lab, you will practice building a threat hunting mind map starting from a single column in the dataset: `Domain / Host`

NOTE: In the sample datasets `Domain / Host` is not included for space savings reasons. We can extract `Domain / Host` from the `URL` field by running the following Bash command.

```
awk -F',' 'BEGIN {OFS=","} { split($8, a, "/"); print a[1] }'
lab_github.csv
```

The output will look something like this:

```
www.msftconnecttest.com
array810.prod.do.dsp.mp.microsoft.com
www.msftconnecttest.com
login.live.com
edge-consumer-static.azureedge.net
edge-cloud-resource-static.azureedge.net
edge.microsoft.com
```

Listing the parsed out domain names from the logs.

Lab Objective

The objective of this lab is to build a visual mind map that helps you generate threat hunting hypotheses from web proxy telemetry.

You will:

- Start with a central Domain / Host node
- Expand outward into multiple analytical dimensions
- Form “what-if” questions that can later be tested with data

This lab focuses on **thinking and hypothesis formation**, not detection or validation.

Tools

You may use **any medium** to create your mind map:

- Miro
- XMind
- Google Slides / Google Drawings
- PowerPoint / Keynote
- A whiteboard or piece of paper (take a photo if needed)

There is no required format, but your final result should visually resemble a **hub-and-spoke mind map**.

Instructions

Step 1: Create the Core Node

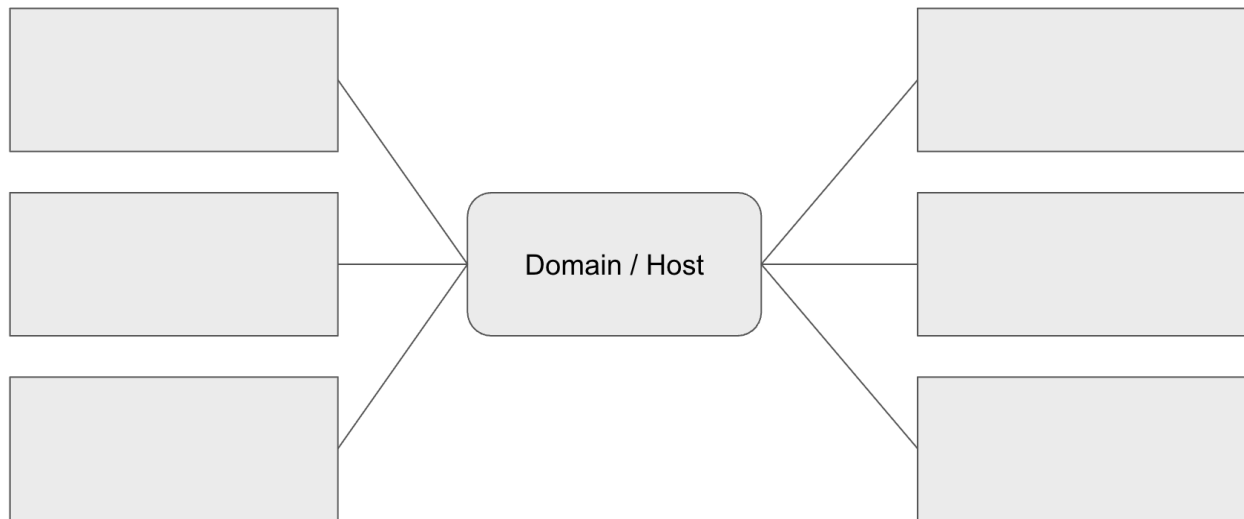
Place the following at the **center** of your mind map:



This represents any external domain or host observed in SWG/ZIA telemetry (e.g., HTTP(S) requests, DNS lookups, downloads).

Step 2: Add Primary Branches

From the central **Domain / Host** node, create **4–6 primary branches**.
You may use the examples below or adapt them based on your experience.



Suggested primary branches:

- Uncommon Domains
- Newly Registered Domains
- Commonly Abused TLDs
- Known Bad or Reputation Signals
- Subdomain Patterns
- Associated Behaviors

Each branch represents a different **analytical lens** you might apply during a hunt.

Step 3: Add Sub-Branches with “What-If” Questions

For **each primary branch**, add **2–3 sub-branches** that explore **what-if scenarios**. These should be written as **questions**, not statements.

The goal is to ask:

“If this were malicious, what might it look like?”

Below are **example prompts** to get you started. You do **not** need to use all of them.

Example Branch Ideas (Use as Inspiration)

Uncommon Domains

- What if the domain has very low global prevalence?
- What if it does not appear in common popularity rankings?
- What if only one user in the environment ever accessed it?

Newly Registered Domains

- What if the domain was registered within the last 30 days?
- What if it appeared for the first time during this session?
- What if multiple users suddenly accessed a new domain?

Commonly Abused TLDs

- What if the domain uses a low-cost or frequently abused TLD?
- What if the TLD is uncommon for our organization's geography?
- What if similar domains with this TLD have been abused historically?

Known Bad or Reputation Signals

- What if threat intelligence tools show weak or emerging signals?
- What if the domain has a small number of detections, but not enough to block?
- What if reputation varies across vendors?

Subdomain Patterns

- What if the subdomain appears randomly generated?
- What if the hostname shows high entropy or unusual character patterns?
- What if subdomains change frequently over a short time window?

Associated Behaviors

- What if the domain is involved in downloads or file transfers?
- What if it is accessed immediately after visiting a trusted site?
- What if it appears alongside unusual user-agent strings or request methods?

Step 4: Connect Ideas and Form Hypotheses

As you build your map:

- Look for **connections between branches**

- Identify combinations that feel more concerning together than alone

For example:

- *A newly registered domain + abused TLD + download behavior*
- *An uncommon domain + DGA-like subdomain + single-user access*

These intersections are where **strong hunt hypotheses** emerge.

Deliverable

Your final deliverable is a **completed mind map** that:

- Has **Domain / Host** as the core node
- Includes **4–6 primary branches**
- Contains **multiple “what-if” sub-branches**
- Demonstrates your reasoning process

There is **no single correct answer**. Two analysts may produce very different—but equally valid—mind maps.

Reflection Questions

Question	Your Answer
Which branch generated the most hypotheses for you?	
Which hypotheses would you prioritize testing first?	
Which ideas would require additional data sources to validate?	
How does mind mapping change how you approach large datasets?	